Zen-Musician:

Symbolic and Audio Music Generation

Hanzo AI Inc.
research@hanzo.ai

Zoo Labs Foundation
research@zoolabs.org

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Abstract

We introduce **Zen-Musician**, a dual-stream transformer architecture for generating musically coherent, multi-instrument compositions in both symbolic (MIDI) and audio (waveform) domains from text descriptions, melody prompts, and structural specifications.

Music generation presents challenges distinct from speech and general audio synthesis: hierarchical temporal structure spanning milliseconds (articulation) to minutes (form), complex harmonic relationships between concurrent voices, stylistic conventions that vary across genres and historical periods, and the subjective nature of musical quality. Prior systems have addressed these challenges either in the symbolic domain (producing MIDI or sheet music) or in the audio domain (producing waveforms), but rarely both, and typically for single instruments or monophonic textures.

Zen-Musician bridges these domains through three architectural innovations. First, a *Hierarchical Music Structure Encoder* (HMSE) represents musical form as a three-level hierarchy — macro-structure (sections, keys), meso-structure (phrases, harmonic progressions), and micro-structure (beats, note onsets) — enabling the model to plan long-range form before filling in local detail. Second, a *Multi-Instrument Orchestration Module* (MIOM) generates instrument-specific token streams in parallel, conditioned on a shared harmonic context, using cross-instrument attention to ensure coherent voicing, register allocation, and rhythmic coordination. Third, a *Dual-Stream Decoder* produces both a symbolic MIDI representation and a high-fidelity audio spectrogram from a shared latent space, enabling the user to obtain editable scores and broadcast-quality audio from a single generation.

Trained on 1.2 million curated compositions spanning 42 genres from the MusicNet, Lakh MIDI, MusicCaps, and a licensed commercial music corpus, Zen-Musician achieves state-of-the-art results on MusicCaps text-to-music generation (FAD 2.14,

CLAP score 0.412), Lakh MIDI symbolic generation (NDB 0.82, note F1 0.87), and a new multi-instrument orchestration benchmark. Human evaluation by 28 professional musicians rates 67% of Zen-Musician outputs as "publishable with minor editing," compared to 34% for the next-best baseline.

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1 Introduction

Music is among the most temporally structured forms of human expression. A compelling musical composition operates simultaneously at multiple time scales: the sub-second dynamics of individual note articulation, the multi-second phrasing of melodic lines, the tens-of-seconds periodicity of harmonic progressions, and the minutes-scale architecture of formal sections (verse, chorus, bridge, development, recapitulation). Generating music that is coherent across all of these scales remains a fundamental challenge for generative models.

Recent advances in audio generation (Borsos et al., 2023; Agostinelli et al., 2023; Copet et al., 2024) have demonstrated the ability to produce plausible-sounding music from text descriptions, but these systems operate entirely in the audio domain and produce monolithic waveforms that cannot be decomposed into individual instrument tracks, edited at the note level, or rendered with alternative timbres. Conversely, symbolic music generation systems (Huang & Yang, 2020; Hsiao et al., 2021; Ren et al., 2020) produce structured MIDI output that is fully editable but requires a separate synthesis step to produce audio, and typically handle only limited instrumentation.

Zen-Musician addresses these limitations through a unified architecture that:

1. **Plans before detailing**: A hierarchical structure encoder represents music at three temporal scales (macro, meso, micro), enabling the model to establish global form, key centers, and section boundaries before generating note-level content.

2. **Orchestrates across instruments**: A multi-instrument module generates separate but harmonically coordinated streams for up to 16 concurrent instruments, using cross-instrument attention to manage voicing, register, and rhythmic interplay.

3. **Outputs both symbolic and audio**: A dual-stream decoder produces synchronized MIDI and mel spectrogram outputs from a shared latent representation, giving users both an editable score and broadcast-ready audio.

1.1 Contributions

- **Hierarchical Music Structure Encoder (HMSE)**: A three-level planning module that establishes macro-structure (form, key plan), meso-structure (phrase boundaries, chord progressions), and micro-structure (rhythm, onset patterns) before note generation.

- **Multi-Instrument Orchestration Module (MIOM)**: Parallel per-instrument generation with cross-instrument attention for harmonic coherence, register avoidance, and rhythmic coordination.

- **Dual-Stream Decoder**: Joint symbolic (MIDI token) and audio (mel spectrogram diffusion) generation from a shared latent, enabling synchronized editable-score and broadcast-audio outputs.

- **OrchBench**: A new benchmark for evaluating multi-instrument orchestration quality, covering 500 scored orchestral excerpts with ground-truth part assignments.
- **State-of-the-art results** on MusicCaps (FAD 2.14), Lakh MIDI (NDB 0.82, note F1 0.87), and OrchBench (orchestration score 0.74).

2 Background and Related Work

2.1 Symbolic Music Generation

Music Transformer (Huang & Yang, 2020) applied relative attention to MIDI event sequences, producing coherent piano performances. Compound Word Transformer (Hsiao et al., 2021) grouped MIDI events into compound tokens, improving structural coherence. MuseNet (Payne, 2019) demonstrated multi-instrument generation via a large GPT-style model. These approaches model music as flat token sequences; the absence of explicit hierarchical structure limits formal coherence over extended durations.

2.2 Audio Music Generation

AudioLM (Borsos et al., 2023) demonstrated high-fidelity audio continuation over SoundStream (Zeghidour et al., 2021) codes. MusicLM (Agostinelli et al., 2023) extended this to text-conditioned generation. MusicGen (Copet et al., 2024) introduced a single-stage model over EnCodec (Défossez et al., 2022) tokens. Stable Audio (Evans et al., 2024) applied latent diffusion to long-form generation. These models produce convincing textures but lack editability and struggle with precise harmonic structure, as audio codecs do not explicitly represent pitch relationships.

2.3 Bridging Symbolic and Audio Domains

MIDI-DDSP (Wu et al., 2022) maps MIDI to audio via differentiable synthesizers. SingSong (Donahue et al., 2023) generates accompaniment audio from vocal input. No prior system jointly generates symbolic and audio outputs from text within a single architecture.

2.4 Music Structure and Orchestration

Lerdahl and Jackendoff's GTTM (Lerdahl & Jackendoff, 1983) formalizes the hierarchical grouping of tonal music. Computational structural analysis (McFee & Ellis, 2017; Nieto et al., 2020) enables automatic section boundary detection. For orchestration, OrchideaSOL (Cella et al., 2020) provides a database for computational orchestration, and deep learning approaches (Crestel & Esling, 2018) have shown promise but remain limited to single-instrument timbre transfer.

3 Tokenization and Representation

3.1 Symbolic Tokenization

Zen-Musician uses a compound token representation for symbolic music based on Hsiao et al. (2021), extended with structural and orchestration tokens. Each musical event is represented as a compound token containing multiple fields:

Definition 3.1 (Compound Music Token)**.** *A compound music token* $\mathbf{t} = (t_{type}, t_{beat}, t_{pitch}, t_{dur}, t_{vel}, t_{inst}, t_{expr})$ *encodes event type (NOTE, CHORD, BAR, PHRASE, SECTION, KEY, TEMPO, DYN), beat position (128th-note resolution), MIDI pitch, duration, velocity, General MIDI instrument program, and expression code (legato, staccato, vibrato, etc.).*

Each field is embedded independently and summed to form the compound token embedding:

$$\mathbf{e}(\mathbf{t}) = \sum_{f \in \text{fields}} \mathbf{W}_f \cdot \text{Embed}_f(t_f) \tag{1}$$

where $\mathbf{W}_f \in \mathbb{R}^{d_{\text{model}} \times d_f}$ are field-specific projection matrices and d_f is the embedding dimension for field f.

3.2 Structural Tokens

We introduce structural tokens encoding hierarchical music structure: `[SECTION:type]` (intro, verse, chorus, bridge, etc.), `[KEY:root/mode]` (e.g., C major), `[TEMPO:bpm]` (quantized to 20 bins from 40–240 BPM), `[PHRASE:start/end]` (phrase boundaries), and `[CHORD:quality]` (48 classes: 12 roots × 4 qualities).

3.3 Audio Tokenization

For the audio stream, we use EnCodec (Défossez et al., 2022) at 24 kHz with 4 codebook levels and a frame rate of 75 Hz, yielding 300 tokens per second of audio. Additionally, we represent audio via log-amplitude mel spectrograms (128 bins, 256-sample hop at 24 kHz, yielding 93.75 Hz frame rate) for the diffusion decoder.

3.4 Vocabulary Statistics

Table 1: Token vocabulary sizes for each representation domain.

Domain	Vocab Size	Tokens per Min (typical)
Symbolic compound tokens	8,192	∼800
Structural tokens	256	∼30
Audio EnCodec (4 levels)	4,096 × 4	18,000
Mel spectrogram (continuous)	—	5,625 frames

4 Model Architecture

4.1 Overview

Zen-Musician consists of four major components operating in a top-down generation pipeline:

1. **Text and Conditioning Encoder**: Encodes text descriptions, optional melody prompts, and structural specifications into conditioning vectors.
2. **Hierarchical Music Structure Encoder (HMSE)**: Plans the global form, key scheme, phrase structure, and rhythmic skeleton.
3. **Multi-Instrument Orchestration Module (MIOM)**: Generates per-instrument symbolic token streams coordinated by the structural plan.
4. **Dual-Stream Decoder**: Produces synchronized MIDI and audio outputs from the shared latent.

4.2 Text and Conditioning Encoder

Text prompts are encoded by a frozen CLAP (Wu et al., 2023) text encoder (512-dim). We support five conditioning modalities: (1) free-form text descriptions, (2) melody prompts (MIDI or hummed audio via CREPE (Kim et al., 2018)), (3) structural specifications (form, bars, key, tempo), (4) instrument lists, and (5) reference audio ($\leq$ 30s, encoded via CLAP audio encoder). All signals are projected to $d = 1024$ and concatenated into a conditioning sequence $\mathbf{C} \in \mathbb{R}^{L_c \times 1024}$.

4.3 Hierarchical Music Structure Encoder

The HMSE generates the structural plan in three stages, each conditioned on the text embedding and the output of the previous stage.

Stage 1: Macro-Structure (Form). An autoregressive transformer ($L_{\text{macro}} = 8$ layers, $d = 512$, 8 heads) generates a sequence of section tokens specifying the formal plan:

$$\mathbf{S}_{\text{macro}} = \text{AR}_{\text{macro}}(\mathbf{C}) = \{s_1, s_2, \ldots, s_K\}, \quad s_k = (\text{type}_k, \text{key}_k, \text{bars}_k, \text{tempo}_k) \tag{2}$$

where K is the number of sections (typically 4–12 for a 3–5 minute composition). For example, a pop song might generate: `[INTRO, C major, 4 bars, 120 BPM]` $\rightarrow$ `[VERSE, C major, 16 bars]` $\rightarrow$ `[CHORUS, F major, 8 bars]` $\rightarrow$ ....

Stage 2: Meso-Structure (Phrases and Harmony). Conditioned on the macro plan, a second transformer ($L_{\text{meso}} = 12$ layers, $d = 768$, 12 heads) generates per-section phrase boundaries and chord progressions:

$$\mathbf{S}_{\text{meso}}^{(k)} = \text{AR}_{\text{meso}}(s_k, \mathbf{C}) = \{(p_j, \text{chord}_j)\}_{j=1}^{J_k} \tag{3}$$

where p_j is a phrase boundary marker and $\text{chord}_j \in \{1, \ldots, 48\}$ is a chord class label at beat resolution. The meso-structure transformer is trained with chord labels extracted by a music information retrieval (MIR) pipeline using madmom (Böck et al., 2016) and Chordino (Mauch & Dixon, 2010).

Stage 3: Micro-Structure (Rhythm and Onset). A third transformer ($L_{\text{micro}} = 8$ layers, $d = 512$, 8 heads) generates a rhythmic skeleton for each phrase: a binary onset grid at 16th-note resolution indicating *when* notes occur, without specifying pitch or instrument:

$$\mathbf{S}_{\text{micro}}^{(k,j)} = \text{AR}_{\text{micro}}(p_j, \text{chord}_j, \mathbf{C}) = \{o_n \in \{0, 1\}\}_{n=1}^{N_j} \tag{4}$$

where N_j is the number of 16th-note positions in phrase j.

This three-stage plan provides the scaffolding that guides note-level generation in the MIOM.

4.4 Multi-Instrument Orchestration Module

Given the structural plan from the HMSE and the instrument list from the conditioning encoder, the MIOM generates compound music tokens for each instrument in parallel.

4.4.1 Per-Instrument Stream

Each instrument $i \in \{1, \ldots, I\}$ has a dedicated autoregressive decoder stream ($L_{\text{inst}} = 16$ layers, $d = 1024$, 16 heads) that generates its token sequence conditioned on the structural plan:

$$\mathbf{T}^{(i)} = \text{AR}_{\text{inst}}^{(i)}(\mathbf{S}_{\text{macro}}, \mathbf{S}_{\text{meso}}, \mathbf{S}_{\text{micro}}, \mathbf{C}) \tag{5}$$

4.4.2 Cross-Instrument Attention

To ensure coherent orchestration, every 4th layer of the per-instrument decoder includes a *cross-instrument attention* layer. For instrument i at layer ℓ, the hidden states attend to the hidden states of all other instruments at the same layer:

$$\mathbf{H}_\ell^{(i)} \leftarrow \mathbf{H}_\ell^{(i)} + \text{CrossAttn}\left(\mathbf{H}_\ell^{(i)},\ \text{Concat}_{j \neq i}\left[\mathbf{H}_\ell^{(j)}\right]\right) \tag{6}$$

Cross-instrument attention serves three purposes: (1) **harmonic coherence**—instruments see each other's pitch choices, avoiding unsanctioned dissonances; (2) **register allocation**—preventing excessive overlap; (3) **rhythmic coordination**—enabling complementary patterns across instruments.

4.4.3 Instrument Embedding

Each instrument is represented by a learned embedding $\mathbf{e}_{\text{inst}}^{(i)} \in \mathbb{R}^{1024}$ that encodes timbre characteristics, typical register, and playing style. This embedding is added to every token embedding in the instrument's stream and is also used to modulate the

cross-attention weights via adaptive layer normalization:

$$\gamma_\ell^{(i)}, \beta_\ell^{(i)} = \mathrm{MLP}(\mathbf{e}_{\text{inst}}^{(i)}) \tag{7}$$

The instrument embeddings are initialized from a pretrained timbre embedding space learned via contrastive learning on the NSynth (Engel et al., 2017) dataset.

4.4.4 Orchestration Constraints

Hard constraints ensure musical validity: pitch range clamping per instrument (e.g., violin: G3–E7), polyphony limits (monophonic instruments produce at most one concurrent note), and playability heuristics enforcing instrument-specific maximum interval and speed constraints learned from training data.

4.5 Dual-Stream Decoder

The Dual-Stream Decoder produces both symbolic MIDI output and high-fidelity audio output from a shared latent representation.

4.5.1 Shared Latent Space

The per-instrument compound token sequences $\{\mathbf{T}^{(i)}\}_{i=1}^{I}$ are processed by a shared latent encoder ($L_{\text{shared}} = 4$ layers, $d = 1024$) that produces a continuous latent representation:

$$\mathbf{Z} = \mathrm{SharedEnc}\left(\mathrm{Interleave}\left(\mathbf{T}^{(1)}, \ldots, \mathbf{T}^{(I)}\right)\right) \in \mathbb{R}^{N_z \times 1024} \tag{8}$$

where `Interleave` merges instrument streams by temporal position (beat-synchronous interleaving) and N_z is the total number of interleaved tokens.

4.5.2 Symbolic Stream (MIDI Output)

The symbolic stream is a direct readout from the MIOM output. The compound tokens are decoded back to standard MIDI events through a deterministic mapping:

$$\mathrm{MIDI} = \mathrm{Decode}_{\text{sym}}\left(\{\mathbf{T}^{(i)}\}_{i=1}^{I}\right) \tag{9}$$

This produces a multi-track MIDI file with one track per instrument, retaining all compound token fields (pitch, duration, velocity, expression).

4.5.3 Audio Stream (Spectrogram Diffusion)

The audio stream uses a conditional diffusion model in mel spectrogram space, similar to the architecture used in Zen-Foley but specialized for music:

$$\hat{\boldsymbol{\epsilon}} = g_\psi(\mathbf{M}_t, t, \mathbf{Z}, \mathbf{C}) \tag{10}$$

where $\mathbf{M}_t$ is the noised mel spectrogram at diffusion step t, $\mathbf{Z}$ is the shared latent from the symbolic generation, and $\mathbf{C}$ is the text/audio conditioning. The denoiser g_ψ is a U-Net with:

- **FiLM conditioning** from $\mathbf{Z}$ at every resolution scale, providing note-level guidance to the spectrogram generation
- **Cross-attention** to the text conditioning $\mathbf{C}$ at the bottleneck
- **Temporal position encoding** via sinusoidal embeddings aligned to the symbolic beat grid

The mel spectrogram is generated at 128 bins, 24 kHz sample rate, 256-sample hop (93.75 Hz frame rate). A fine-tuned HiFi-GAN vocoder (Kong et al., 2020) converts the mel spectrogram to waveform.

4.5.4 Consistency Loss

To ensure that the symbolic and audio streams remain synchronized, we apply a *consistency loss* during training that aligns the note onsets in the symbolic stream with energy peaks in the generated spectrogram:

$$\mathcal{L}_{\text{consist}} = \sum_{i,n} \left\| \sigma\left(\mathbf{M}^{\text{gen}}_{t_n^{(i)}}\right) - \sigma\left(\mathbf{M}^{\text{real}}_{t_n^{(i)}}\right) \right\|_1 \tag{11}$$

where $t_n^{(i)}$ are the onset times of note n in instrument i and σ is a soft energy envelope extractor.

4.6 Model Configurations

Table 2: Zen-Musician component specifications.

Component	**Specification**	**Parameters**
Text/Conditioning Encoder	CLAP (frozen) + projection	151M (frozen) + 8M
HMSE: Macro Transformer	8 layers, $d = 512$, 8 heads	42M
HMSE: Meso Transformer	12 layers, $d = 768$, 12 heads	108M
HMSE: Micro Transformer	8 layers, $d = 512$, 8 heads	42M
MIOM: Per-Instrument Decoder	16 layers, $d = 1024$, 16 heads	680M
MIOM: Cross-Instrument Attn	Every 4th layer	(included above)
Shared Latent Encoder	4 layers, $d = 1024$	67M
Audio Stream: U-Net Denoiser	4 scales, FiLM conditioning	310M
HiFi-GAN Vocoder	Fine-tuned on music	14M
Total (trainable)		**1.27B**
Max instruments	16 concurrent	
Max duration	5 minutes (symbolic), 3 min (audio)	
Audio output	24 kHz stereo, 128 mel bins	

5 Training

5.1 Training Data

Table 3: Training corpus statistics.

Source	Tracks	Hours	Genres	Type
Lakh MIDI Dataset (Raffel, 2016)	176,000	—	38	Symbolic
MusicNet (Thickstun et al., 2017)	330	34	4	Symbolic + Audi
MusicCaps (Agostinelli et al., 2023)	5,500	28	42	Audio + Text
Free Music Archive (FMA) (Defferrard et al., 2017)	106,000	8,800	42	Audio
Licensed Commercial Corpus	920,000	74,000	42	Symbolic + Audi
Total	**1,207,830**	**82,862**	42	Mixed

Processing Pipeline. Raw MIDI files are tokenized into compound tokens; audio files are converted to mel spectrograms. Audio-only data (FMA) is annotated via MT3 (Gardner et al., 2022) automatic transcription. Tracks without text are captioned using a music tagging model followed by LLM tag-to-sentence conversion (averaging 32 words). Quality filtering retains tracks with SNR > 20 dB, AMT note F1 > 0.7, chord confidence > 0.6, and genre-consistent section boundaries. After filtering, 1.2M tracks remain.

5.2 Training Protocol

Training proceeds in four stages:

Stage 1: HMSE Pre-training (7 days, 64 A100 GPUs). The three HMSE transformers are trained jointly on structural annotations extracted from the full corpus. The macro transformer is trained to predict section sequences from text; the meso transformer predicts chord progressions given sections; the micro transformer predicts onset patterns given phrases and chords.

Loss:

$$\mathcal{L}_{\text{HMSE}} = \mathcal{L}^{\text{CE}}_{\text{macro}} + \mathcal{L}^{\text{CE}}_{\text{meso}} + \mathcal{L}^{\text{BCE}}_{\text{micro}} \tag{12}$$

Batch size 256, learning rate 3×10^{-4}, AdamW with cosine decay.

Stage 2: MIOM Training (21 days, 128 A100 GPUs). The per-instrument decoders and cross-instrument attention are trained on compound token sequences from the symbolic data (Lakh MIDI + Licensed Commercial Corpus), conditioned on HMSE structural plans.

Loss (per-instrument cross-entropy):

$$\mathcal{L}_{\text{MIOM}} = \sum_{i=1}^{I} \sum_{n=1}^{N_i} -\log p_\theta(\mathbf{t}_n^{(i)} | \mathbf{t}_{<n}^{(i)}, \mathbf{H}^{(\setminus i)}, \mathbf{S}, \mathbf{C}) \tag{13}$$

where $\mathbf{H}^{(\setminus i)}$ denotes the cross-instrument context from all other instruments. Batch size 128, learning rate 1×10^{-4}.

Stage 3: Audio Stream Training (14 days, 128 A100 GPUs). The spectrogram diffusion U-Net is trained on paired symbolic-audio data from the Licensed Commercial Corpus and MusicNet. The HMSE and MIOM are frozen.

Loss:

$$\mathcal{L}_{\text{audio}} = \mathbb{E}_{\mathbf{M}_0, t, \boldsymbol{\epsilon}} \left[\|\boldsymbol{\epsilon} - g_\psi(\mathbf{M}_t, t, \mathbf{Z}, \mathbf{C})\|_2^2 \right] + \lambda_c \mathcal{L}_{\text{consist}} \tag{14}$$

with $\lambda_c = 0.1$. Batch size 64, learning rate 5×10^{-5}, $T_D = 1000$ diffusion steps with linear noise schedule.

Stage 4: End-to-End Fine-tuning (5 days, 128 A100 GPUs). All components are jointly fine-tuned on paired text-symbolic-audio data with a combined loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{HMSE}} + \mathcal{L}_{\text{MIOM}} + \mathcal{L}_{\text{audio}} + \lambda_c \mathcal{L}_{\text{consist}} \tag{15}$$

Learning rate 2×10^{-5}, batch size 32. End-to-end fine-tuning improves the alignment between structural plan, symbolic output, and audio output.

5.3 Classifier-Free Guidance

Text conditioning is dropped with probability 10%, melody conditioning with 20%, and structural specification with 30% during training. At inference, we use guidance scales:

$$\hat{\mathbf{o}} = \mathbf{o}_{\text{uncond}} + w_{\text{text}}(\mathbf{o}_{\text{text}} - \mathbf{o}_{\text{uncond}}) + w_{\text{melody}}(\mathbf{o}_{\text{melody}} - \mathbf{o}_{\text{uncond}}) \tag{16}$$

with default values $w_{\text{text}} = 3.5$ and $w_{\text{melody}} = 2.0$.

5.4 Inference Algorithm

Algorithm 1 Zen-Musician Inference

Input: Text prompt $\mathbf{c}_{\text{text}}$, optional melody $\mathbf{c}_{\text{mel}}$, optional structure spec, instrument list
Input: Audio diffusion steps $S = 50$
Encode conditioning: $\mathbf{C} = \text{CondEnc}(\mathbf{c}_{\text{text}}, \mathbf{c}_{\text{mel}}, \ldots)$
— **Stage 1: Plan Structure** —
$\mathbf{S}_{\text{macro}} = \text{HMSE.macro}(\mathbf{C})$
$\mathbf{S}_{\text{meso}} = \text{HMSE.meso}(\mathbf{S}_{\text{macro}}, \mathbf{C})$
$\mathbf{S}_{\text{micro}} = \text{HMSE.micro}(\mathbf{S}_{\text{meso}}, \mathbf{C})$
— **Stage 2: Generate Per-Instrument Tokens** —
for each instrument i **in parallel do**
    $\mathbf{T}^{(i)} = \text{MIOM.generate}^{(i)}(\mathbf{S}_*, \mathbf{C})$ ▷ with cross-instrument attention
end for
— **Stage 3: Output Symbolic** —
$\text{MIDI} = \text{Decode}_{\text{sym}}(\{\mathbf{T}^{(i)}\})$
— **Stage 4: Generate Audio** —
$\mathbf{Z} = \text{SharedEnc}(\text{Interleave}(\mathbf{T}^{(1)}, \ldots, \mathbf{T}^{(I)}))$
$\mathbf{M}_S \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $s = S$ **down to** 1 **do**
    $\hat{\boldsymbol{\epsilon}} = g_\psi(\mathbf{M}_s, s, \mathbf{Z}, \mathbf{C})$ with CFG
    $\mathbf{M}_{s-1} = \text{DDIM_step}(\mathbf{M}_s, \hat{\boldsymbol{\epsilon}}, s)$
end for
$\hat{\mathbf{a}} = \text{HiFi-GAN}(\mathbf{M}_0)$
return MIDI, $\hat{\mathbf{a}}$

6 Evaluation

We evaluate Zen-Musician on three dimensions: text-to-audio generation quality, symbolic music generation quality, and multi-instrument orchestration quality.

6.1 Evaluation Metrics

- **FAD** (Fréchet Audio Distance) (Kilgour et al., 2019): Distributional distance in VGGish embedding space; lower is better.
- **CLAP Score**: Cosine similarity between CLAP embeddings of text and generated audio; higher is better.
- **KL Divergence**: KL divergence between generated and real audio event class distributions.
- **NDB** (Number of statistically Different Bins) (Richardson & Weiss, 2018): Measures coverage of the real data distribution by the generated distribution in symbolic

token space; lower is better.

- **Note F1**: Precision and recall of generated notes compared to ground truth MIDI (onset tolerance 50 ms, pitch tolerance 0 semitones, duration tolerance 50 ms).
- **Orchestration Score**: A composite metric for multi-instrument quality measuring harmonic consistency, register separation, and rhythmic complementarity.

6.2 Text-to-Audio Generation (MusicCaps)

Table 4: Text-to-audio evaluation on MusicCaps test set ($n = 400$). Best results in bold.

Model	FAD ↓	CLAP Score ↑	KL Div. ↓
MusicLM (Agostinelli et al., 2023)	4.72	0.318	1.82
MusicGen-Large (Copet et al., 2024)	3.41	0.356	1.43
Stable Audio 2.0 (Evans et al., 2024)	2.87	0.381	1.21
JEN-1 (Li et al., 2024)	3.18	0.342	1.54
Zen-Musician (audio only)	2.42	0.398	1.04
Zen-Musician (full pipeline)	**2.14**	**0.412**	**0.91**

Zen-Musician achieves an FAD of 2.14, improving over Stable Audio 2.0 by 25%. The full pipeline (including symbolic generation as an intermediate step) outperforms the audio-only variant by 0.28 FAD, confirming that the symbolic intermediate representation provides useful structural guidance to the audio decoder.

6.3 MusicCaps Genre Breakdown

Table 5: FAD by genre on MusicCaps (selected genres with $n \geq 20$ test samples).

Genre	MusicGen	Stable Audio	Zen-Musician	Improv.
Pop/Rock	3.12	2.64	**1.98**	−25.0%
Classical	4.87	3.71	**2.41**	−35.0%
Jazz	4.21	3.28	**2.18**	−33.5%
Electronic	2.81	2.33	**1.82**	−21.9%
Hip-Hop/R&B	3.58	2.91	**2.34**	−19.6%
World/Folk	4.94	3.84	**2.87**	−25.3%
Overall	3.41	2.87	**2.14**	−25.4%

Improvements are largest for Classical (−35.0%) and Jazz (−33.5%), where complex orchestration makes MIOM and HMSE most impactful. Electronic music shows the smallest gain.

6.4 Symbolic Music Generation (Lakh MIDI)

Table 6: Symbolic music generation evaluation on Lakh MIDI test set ($n = 2{,}000$).

Model	NDB ↓	Note F1 ↑	Pitch Entropy	Duration Var.
Music Transformer (Huang & Yang, 2020)	1.28	0.71	3.42	0.84
Compound Word (Hsiao et al., 2021)	1.14	0.76	3.51	0.89
MuseNet (Payne, 2019)	1.02	0.79	3.64	0.91
Zen-Musician (symbolic only)	**0.82**	**0.87**	3.71	0.94

Zen-Musician achieves an NDB of 0.82 (lower is better), indicating that the generated distribution covers 82% of the real data's statistical bins without over-concentrating. Note F1 of 0.87 indicates high accuracy in reproducing the pitch, timing, and duration patterns of the target genre. Pitch entropy (3.71 vs. 3.75 for real data) and duration variability (0.94 vs. 0.97 for real data) are close to ground truth, indicating natural diversity without mode collapse.

6.5 Multi-Instrument Orchestration (OrchBench)

Table 7: OrchBench evaluation ($n = 500$ scored orchestral excerpts with 4–12 instruments).

Model	Harm. Consist. ↑	Register Sep. ↑	Rhythm Comp. ↑	Orch. Score ↑	Timbral Diversity ↑
MuseNet	0.61	0.54	0.48	0.54	0.58
MusicGen (mono)	—	—	—	—	0.42
Zen-Musician	**0.81**	**0.72**	**0.68**	**0.74**	**0.71**

The orchestration score (composite of harmonic consistency, register separation, and rhythmic complementarity) of 0.74 substantially exceeds MuseNet's 0.54, demonstrating the effectiveness of the cross-instrument attention mechanism. Register separation (0.72 vs. 0.54) is particularly improved, confirming that the MIOM's mutual awareness prevents instruments from crowding the same pitch range.

6.6 Human Evaluation

We conducted a listening study with 28 professional musicians (mean 14 years of training, including 8 orchestral musicians, 6 jazz musicians, 8 pop/electronic producers, and 6 composers). Participants rated 40 randomly assigned compositions on a 5-point Likert scale across four dimensions and provided an overall publishability judgment.

Table 8: Human evaluation by professional musicians ($n = 28$, 1,120 ratings total).

Dimension	MusicGen	Stable Audio	Zen-Musician	Real Music
Melodic Quality (1–5)	2.84	3.12	**3.87**	4.52
Harmonic Quality (1–5)	2.71	3.04	**3.94**	4.61
Rhythmic Quality (1–5)	3.21	3.38	**3.82**	4.44
Orchestration (1–5)	2.48	2.81	**3.91**	4.58
Overall Quality (1–5)	2.74	3.08	**3.88**	4.53
"Publishable (minor edit)"	14.2%	22.8%	**67.1%**	94.3%
"Needs major revision"	41.4%	34.1%	**18.7%**	3.2%
"Unacceptable"	44.4%	43.1%	**14.2%**	2.5%

Zen-Musician's overall quality of 3.88/5.00 is within 0.65 of real music. The orchestration dimension shows the largest baseline gap (+1.10 over Stable Audio). Publishability rate of 67.1% nearly triples that of Stable Audio (22.8%).

6.7 Structural Coherence Analysis

Table 9: Structural coherence measured by section boundary detection accuracy and harmonic progression quality on 3-minute generations.

Model	Section F1 ↑	Chord Acc. ↑	Key Stability ↑
MusicGen-Large	0.31	0.42	0.68
Stable Audio 2.0	0.38	0.48	0.72
Zen-Musician (no HMSE)	0.44	0.57	0.78
Zen-Musician (full)	**0.71**	**0.73**	**0.91**

The HMSE improves section boundary F1 from 0.44 to 0.71 and chord accuracy from 0.57 to 0.73, confirming that explicit hierarchical planning substantially improves structural coherence. Key stability (the fraction of time the generated music remains in a perceptually stable tonal center) increases from 0.78 to 0.91.

7 Ablation Studies

Table 10: Ablation study on MusicCaps test set (FAD, CLAP) and Lakh MIDI (NDB, Note F1).

Configuration	**FAD ↓**	**CLAP ↑**	**NDB ↓**	**Note F1 ↑**
Full model	**2.14**	**0.412**	**0.82**	**0.87**
– HMSE (flat generation)	3.01	0.371	1.04	0.78
– Cross-instrument attention	2.68	0.391	0.91	0.82
– Audio stream (symbolic only)	—	—	0.82	0.87
– Symbolic stream (audio only)	2.42	0.398	—	—
– Consistency loss	2.31	0.404	0.84	0.86
– Instrument embeddings	2.52	0.394	0.89	0.84
– Orchestration constraints	2.38	0.407	0.86	0.85
– End-to-end fine-tuning	2.47	0.396	0.88	0.85
– Licensed data (public only)	3.24	0.358	0.98	0.81
Macro only (no meso/micro)	2.78	0.384	0.96	0.81
Macro + meso (no micro)	2.41	0.401	0.88	0.85

Key observations: (1) **HMSE** is the most impactful component (-0.87 FAD, -0.22 NDB), with each level contributing incrementally (macro → meso: -0.37; meso → micro: -0.27). (2) **Cross-instrument attention** provides -0.54 FAD, with the largest effect on Classical (-0.91) and Jazz (-0.72). (3) The **symbolic intermediate representation** improves audio quality by -0.28 FAD over audio-only generation. (4) **Licensed data** is critical (-1.10 FAD without it). (5) **End-to-end fine-tuning** adds a modest -0.33 FAD by aligning component interfaces.

7.1 Guidance Scale Analysis

Table 11: Effect of text guidance scale on MusicCaps evaluation.

w_{text}	**FAD ↓**	**CLAP ↑**	**Diversity ↑**
1.0 (no guidance)	3.41	0.312	**0.94**
2.0	2.58	0.378	0.88
3.5 (default)	**2.14**	**0.412**	0.81
5.0	2.21	0.408	0.72
7.0	2.54	0.401	0.61

The optimal text guidance scale of 3.5 balances text adherence (CLAP score) with diversity and audio quality (FAD). Higher guidance scales improve text alignment but reduce diversity and eventually degrade audio quality due to over-concentration.

7.2 Scaling Study

Quality improves log-linearly with MIOM parameter count: Zen-Musician-S (170M MIOM): FAD 3.21, NDB 1.08; -M (340M): FAD 2.72, NDB 0.94; -L (680M, reported): FAD 2.14, NDB 0.82; -XL (1.4B, unreleased): FAD 1.98, NDB 0.78. Further scaling is expected to yield continued improvements.

8 Computational Analysis

Table 12: Inference latency for generating a 3-minute composition on a single A100-80GB.

Stage	**Time (s)**	**Memory (GB)**
Conditioning encoding	0.3	1.2
HMSE (3 stages)	2.1	3.4
MIOM (8 instruments, parallel)	18.7	24.1
Shared latent encoding	1.4	4.8
Audio diffusion (50 steps)	42.3	31.2
HiFi-GAN vocoding	1.8	2.1
Total (symbolic only)	**22.5**	24.1
Total (symbolic + audio)	**66.6**	31.2

Symbolic-only generation (MIDI output) takes 22.5 seconds for a 3-minute composition, approximately 8× faster than real-time. With audio generation, the total is 66.6 seconds (2.7× faster than real-time). The audio diffusion stage dominates latency; reducing diffusion steps from 50 to 20 decreases total time to 34.1 seconds with only 0.18 FAD degradation.

9 Applications

Zen-Musician's dual-stream output supports several application domains:

- **Music production**: Producers generate compositions from text or melody seeds, edit the MIDI in a DAW (Ableton, Logic Pro), and re-render through the audio stream or alternative synthesizers.
- **Film and game scoring**: The structural specification input enables composers to specify exact section timings, instrumentation changes, and mood transitions aligned with narrative structure.
- **Music education**: The symbolic output with HMSE structural annotations makes compositional "reasoning" transparent, enabling students to analyze form, harmony, and orchestration decisions.
- **Accessibility**: Natural language descriptions enable individuals without formal training to create multi-instrument compositions.

10 Limitations

- **Expressive performance**: The model approximates but does not fully capture rubato, dynamic shaping, and idiosyncratic performer articulation.
- **Audio fidelity**: 24 kHz output (12 kHz bandwidth) is adequate for most content but insufficient for high-frequency detail in cymbals and electronic synthesis. Upgrading to 48 kHz is planned.
- **Lyrics and vocals**: Instrumental music only; vocal synthesis via Zen-Scribe is not yet integrated.
- **Copyright**: The model may generate passages resembling copyrighted compositions. Post-generation plagiarism checking is recommended for commercial use.
- **Genre imbalance**: Western tonal music is overrepresented; non-Western genres show noticeably lower quality.
- **Inference cost**: 66.6 seconds for 3 minutes of music is acceptable for offline production but too slow for real-time use.
- **Duration**: Reliable up to 5 minutes (symbolic) and 3 minutes (audio); quality degrades beyond 5 minutes.

11 Ethical Considerations

Music generation raises ethical concerns around musician displacement, attribution, training data compensation, and cultural sensitivity. We position Zen-Musician as a tool for augmenting human creativity: the editable MIDI output facilitates human-AI collaboration rather than replacement. Our Licensed Commercial Corpus includes licensing fees paid to rights holders. We recommend clear AI-generated content labeling and flag culturally specific genres in the user interface to encourage informed use. Non-Western musical traditions deserve respectful treatment; generation in these styles without cultural understanding risks trivialization.

12 Conclusion

We have presented Zen-Musician, a 1.27B-parameter dual-stream transformer for generating multi-instrument music in both symbolic (MIDI) and audio domains from text descriptions, melody prompts, and structural specifications. The HMSE, MIOM, and Dual-Stream Decoder address long-range structural coherence, multi-voice coordination, and cross-domain output.

Zen-Musician achieves state-of-the-art results on MusicCaps (FAD 2.14), Lakh MIDI (NDB 0.82, Note F1 0.87), and OrchBench (orchestration score 0.74). Human evaluation rates 67% of outputs as publishable with minor editing. The hierarchical planning

approach is the most impactful contribution (-0.87 FAD), confirming that music benefits from explicit structural planning before detailed generation.

Future work will address vocal synthesis integration, 48 kHz audio, non-Western music traditions, real-time interactive generation, and scaling to longer compositions.

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